

On Large Deviations in the ABC Conjecture and Prime Factor Constraints

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Zusammenfassung

We study large deviations in the ABC conjecture via the quantity

$$\Delta(a, b, c) := \log c - \log \operatorname{rad}(abc).$$

Numerical experiments indicate a strong negative correlation between Δ and the number $\omega(abc)$ of distinct prime factors.

We formulate a structural hypothesis stating that large values of Δ can only occur when the prime support is small, and discuss how this is consistent with known results on S-unit equations and linear forms in logarithms.

1 Introduction

The ABC conjecture states that for coprime integers a, b with $a + b = c$, and for every $\varepsilon > 0$, there exists K_ε such that

$$c \leq K_\varepsilon \cdot \operatorname{rad}(abc)^{1+\varepsilon}.$$

We define

$$\Delta(a, b, c) := \log c - \log \operatorname{rad}(abc).$$

The conjecture implies that Δ cannot grow significantly.

2 Numerical Evidence

Computations for coprime pairs (a, b) with $a, b \leq N$ (up to several million) show:

- Δ is typically negative,
- large values of Δ are rare,
- large Δ occurs only when $\omega(abc)$ is small.

These observations motivate the following hypothesis.

3 Structural Hypothesis

Lemma 1 (Structural Constraint). *For every $\delta > 0$, there exists a constant $K(\delta)$ such that*

$$\Delta(a, b, c) \geq \delta \implies \omega(abc) \leq K(\delta).$$

Remark 1. *We do not prove this lemma; it is supported by numerical evidence and consistency with known diophantine constraints.*

4 Connection to S-Unit Equations

Let

$$S := \{p : p \mid abc\}, \quad s = |S|.$$

Setting

$$x = \frac{a}{c}, \quad y = \frac{b}{c},$$

we obtain

$$x + y = 1,$$

where x, y are S-units.

By classical results (see [2]), the number of such solutions is finite for fixed S .

5 Baker-Type Bounds

Results on linear forms in logarithms (see [1]) yield bounds of the form

$$\log c \leq C(s) \cdot P \cdot (\log P)^s,$$

where P is the largest prime divisor of abc and $C(s)$ grows at most exponentially in s .

6 Heuristic Consequences

Using the prime number theorem,

$$\log \text{rad}(abc) \gtrsim s \log s.$$

Thus, for large s , the radical term dominates, and Δ cannot be large.

7 Quantitative Behaviour

The above bounds suggest the existence of a function $K(\delta)$ such that

$$\Delta \geq \delta \implies \omega(abc) \leq K(\delta),$$

with $K(\delta)$ growing slowly as $\delta \rightarrow 0$.

8 Discussion

This perspective suggests that large deviations in the ABC conjecture are structurally constrained and correspond to highly restricted prime factor configurations.

9 Outlook

A possible route toward the ABC conjecture is to rigorously establish the structural constraint or derive it from known diophantine estimates.

Literatur

- [1] A. Baker and G. Wüstholz, *Logarithmic Forms and Diophantine Geometry*, Cambridge University Press, 2007.
- [2] J.-H. Evertse and K. Győry, *Unit Equations in Diophantine Number Theory*, Cambridge University Press, 2015.